Confidence Intervals (CI)

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VI. CI for the Population Proportion

I. Size of Difference between Two Means

t Test to evaluate the null hypothesis of "no difference" between two groups

$$H_0 : \mu_1 = \mu_2$$

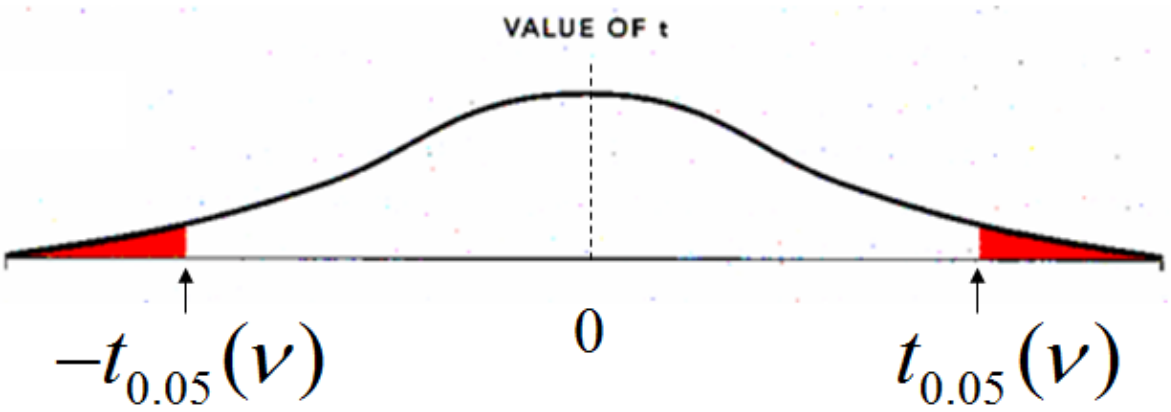
$$t = \frac{\bar{x}_1 - \bar{x}_2}{s_{(\bar{x}_1 - \bar{x}_2)}} \quad \text{where} \quad s_{(\bar{x}_1 - \bar{x}_2)} = \sqrt{\frac{s^2}{n_1} + \frac{s^2}{n_2}} \quad \text{and} \quad s^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

$$\nu = n_1 + n_2 - 2$$

if $|t| > t_\alpha(\nu)$ then reject null hypothesis with $P < \alpha$

t-Distribution Under H_0

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s_{(\bar{x}_1 - \bar{x}_2)}}$$



95% of the values of t fall in the interval: $-t_{0.05}(\nu)$ to $t_{0.05}(\nu)$

$$-t_{0.05}(\nu) < t < t_{0.05}(\nu)$$

t-Distribution When $\mu_1 \neq \mu_2$

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{s_{(\bar{x}_1 - \bar{x}_2)}} \quad \Rightarrow \quad -t_{0.05}(\nu) < t < t_{0.05}(\nu)$$

$$-t_{0.05}(\nu) < \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{s_{(\bar{x}_1 - \bar{x}_2)}} < t_{0.05}(\nu)$$

Solve for $(\mu_1 - \mu_2)$, the true difference of means of the populations:

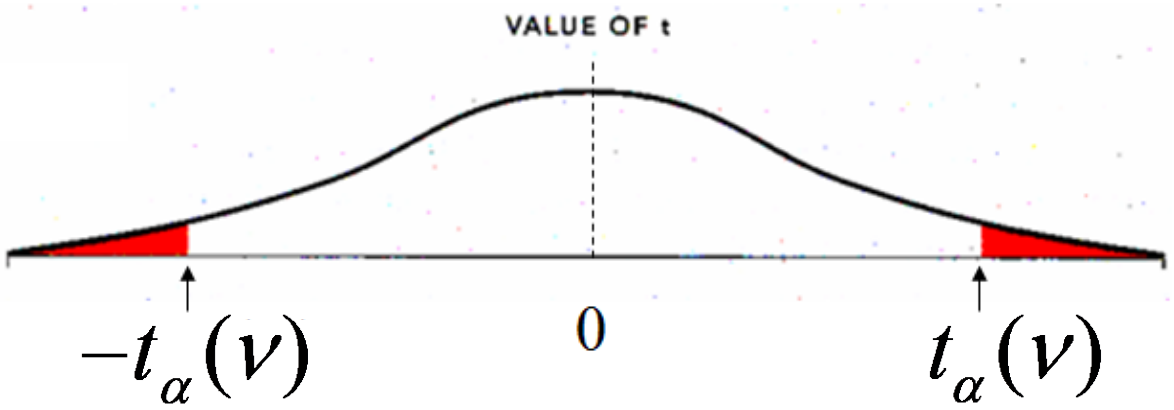
$$(\bar{x}_1 - \bar{x}_2) - t_{0.05}(\nu) \cdot s_{(\bar{x}_1 - \bar{x}_2)} < \mu_1 - \mu_2 < (\bar{x}_1 - \bar{x}_2) + t_{0.05}(\nu) \cdot s_{(\bar{x}_1 - \bar{x}_2)}$$

95% CI for $(\mu_1 - \mu_2)$

Confidence Interval for $(\mu_1 - \mu_2)$

In General: 100·(1-α)% CI for $(\mu_1 - \mu_2)$

$$\left(\bar{x}_1 - \bar{x}_2\right) - t_\alpha(\nu) \cdot s_{\left(\bar{x}_1 - \bar{x}_2\right)} < \mu_1 - \mu_2 < \left(\bar{x}_1 - \bar{x}_2\right) + t_\alpha(\nu) \cdot s_{\left(\bar{x}_1 - \bar{x}_2\right)}$$



$(\mu_1 - \mu_2)$ Confidence Interval Example

Urine Production with Placebo versus Diuretic

$$\bar{x}_{dr} = 1400 \text{ mL/day} \quad s_{dr} = 245 \quad n = 10$$

$$\bar{x}_{pla} = 1180 \text{ mL/day} \quad s_{pla} = 144 \quad n = 10$$

$$s^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2} = (201)^2$$

$$\nu = n_1 + n_2 - 2 = 18$$

$$s_{(\bar{x}_{dr} - \bar{x}_{pla})} = \sqrt{\frac{s^2}{n_{dr}} + \frac{s^2}{n_{pla}}} = 89.9 \text{ mL/day}$$

$$(\bar{x}_{dr} - \bar{x}_{pla}) - t_{0.05}(18) \cdot s_{(\bar{x}_{dr} - \bar{x}_{pla})} < \mu_{dr} - \mu_{pla} < (\bar{x}_{dr} - \bar{x}_{pla}) + t_{0.05}(18) \cdot s_{(\bar{x}_{dr} - \bar{x}_{pla})}$$

$$220 - 2.101(89.9) < \mu_{dr} - \mu_{pla} < 220 + 2.101(89.9)$$

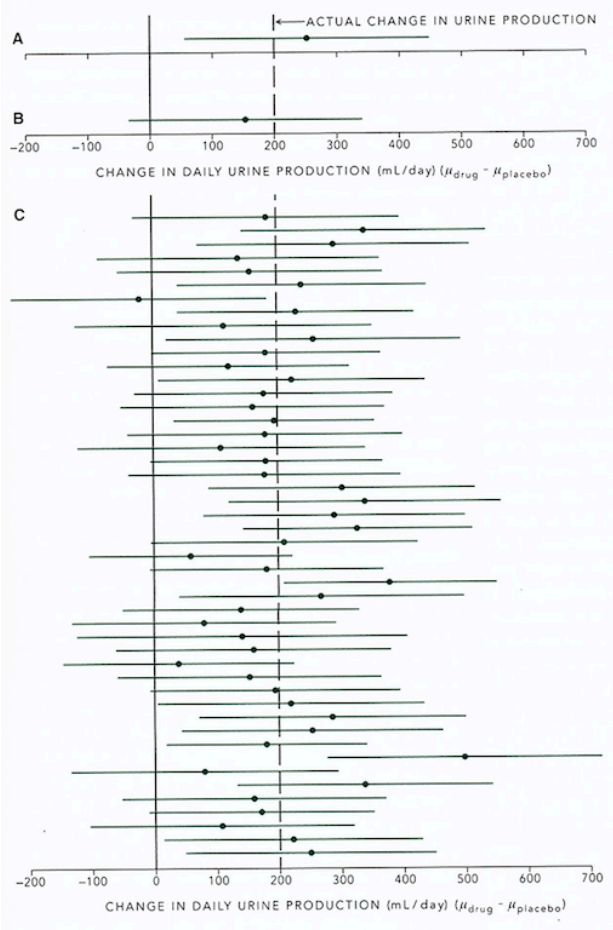
95% CI

$$31 \text{ mL/day} < \mu_{dr} - \mu_{pla} < 409 \text{ mL/day}$$

II. What does Confidence Interval mean?

- 95% of all possible *95% Confidence Intervals* will contain the <u>true</u> difference of the means

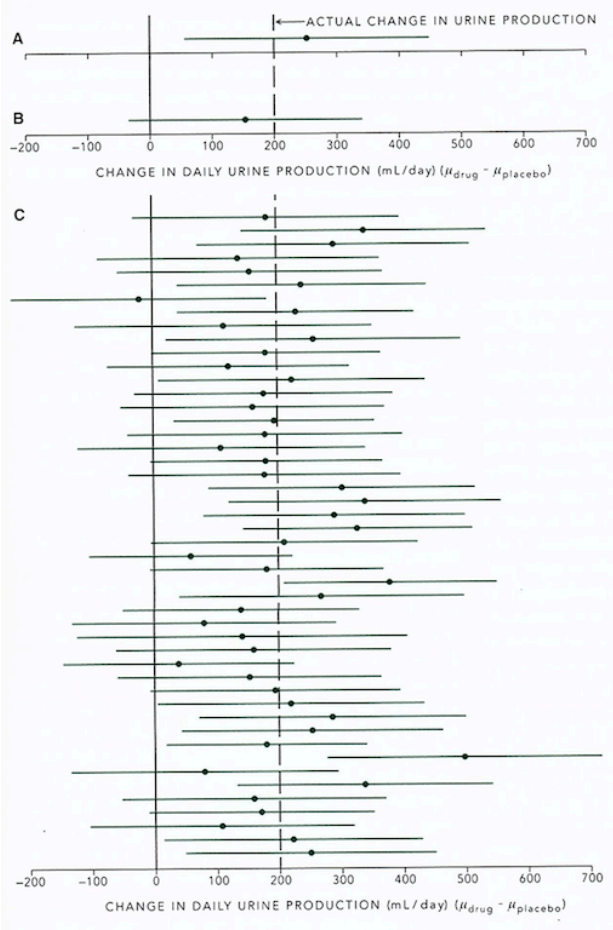
- There is a 95% chance that the *95% Confidence Interval* will contain the true difference of the means



III. Use of CI to Test Null Hypothesis

- If the *100•(1-α)% Confidence Interval* includes 0.0, do not reject H_0

- If the *100•(1-α)% Confidence Interval* does not include 0.0, reject H_0 with $P < \alpha$



IV. Confidence Interval for the Population Mean

Hypothesis test for population mean:

$$H_0 : \mu = \mu_0 \qquad H_1 : \mu \neq \mu_0$$

$$t = \frac{\textit{sample mean} - \textit{population mean}}{\textit{standard error of mean}}$$

$$t = \frac{\bar{x} - \mu}{s_{\bar{x}}} \qquad s_{\bar{x}} = \frac{s}{\sqrt{n}} \qquad \nu = n - 1$$

$$-t_\alpha(\nu) < t < t_\alpha(\nu)$$

$$\bar{x} - t_\alpha(\nu) \cdot s_{\bar{x}} < \mu < \bar{x} + t_\alpha(\nu) \cdot s_{\bar{x}}$$

Therefore, can construct a CI for population mean

Population Mean CI Example

95% CI for μ_{HR} for BME 423 students

$\bar{x} = 67$ beats/min $\quad s = 9$ beats/min $\quad n = 10$

$$s_{\bar{x}} = \frac{s}{\sqrt{n}} = \frac{9}{\sqrt{10}} = 2.85 \quad \nu = n - 1 = 9 \quad t_{0.05}(9) = 2.262$$

$$\bar{x} - t_{\alpha}(\nu) \cdot s_{\bar{x}} < \mu < \bar{x} + t_{\alpha}(\nu) \cdot s_{\bar{x}}$$

$$60.6 < \mu_{HR,423} < 73.4 \quad \text{beats/min}$$

V. Size of Difference between two Proportions

Hypothesis test for two proportions:

$$H_0 : p_1 = p_2$$

$$z = \frac{\left(\hat{p}_1 - \hat{p}_2\right) - \left(p_1 - p_2\right)}{s_{\left(\hat{p}_1 - \hat{p}_2\right)}} \qquad z \sim N(0,1) = t(\nu \to \infty)$$

Remember:

$$\hat{p} = \frac{m_1 + m_2}{n_1 + n_2} \qquad s = \sqrt{\hat{p}(1-\hat{p})} \qquad s_{\hat{p}_1 - \hat{p}_2} = \sqrt{\frac{s^2}{n_1} + \frac{s^2}{n_2}}$$

95% of the values of z fall in the interval: $-z_{0.05}$ to $+z_{0.05}$

$$-t_{0.05}(\infty) = -z_{0.05} < z < z_{0.05} = t_{0.05}(\infty)$$

In General, 100·(1-α)% CI for $\left(p_1 - p_2\right)$, the true difference of proportions of the populations is:

$$\left(\hat{p}_1 - \hat{p}_2\right) - z_\alpha \cdot s_{\left(\hat{p}_1 - \hat{p}_2\right)} < p_1 - p_2 < \left(\hat{p}_1 - \hat{p}_2\right) + z_\alpha \cdot s_{\left(\hat{p}_1 - \hat{p}_2\right)}$$

$(p_1 - p_2)$ Confidence Interval Example

Incidence of Thrombosis with Aspirin versus Placebo

Placebo $\hat{p}_1 = \frac{18}{25} = 0.72 \qquad n_1 = 25$

Aspirin $\hat{p}_2 = \frac{6}{19} = 0.32 \qquad n_2 = 19$

$s_{(\hat{p}_1 - \hat{p}_2)} = 0.15$

$$\left(\hat{p}_1 - \hat{p}_2\right) - z_{0.05} \cdot s_{(\hat{p}_1 - \hat{p}_2)} < p_1 - p_2 < \left(\hat{p}_1 - \hat{p}_2\right) + z_{0.05} \cdot s_{(\hat{p}_1 - \hat{p}_2)}$$

$$0.40 - 1.96(0.15) < p_1 - p_2 < 0.40 + 1.96(0.15)$$

$$0.11 < p_1 - p_2 < 0.69$$

95% confident that aspirin reduces incidence of thrombosis between 11% and 69% compared to placebo

VI. CI for the Population Proportion

Hypothesis test for population proportion:

$$H_0 : p = p_0$$

$$z = \frac{sample\ proportion - true\ proportion}{standard\ error\ of\ \ proportion}$$

$$z = \frac{\hat{p} - p}{s_{\hat{p}}} \qquad s_{\hat{p}} = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

$$\hat{p} - z_\alpha \cdot s_{\hat{p}} < p < \hat{p} + z_\alpha \cdot s_{\hat{p}}$$

Therefore, can construct a CI for true population proportion

Example CI for the Population Proportion

One question on a survey of college students was "Did you take a non-prescribed stimulant in the last year?" Of the 500 students who responded, 177 said "yes."

Is the proportion of students who took stimulants non-medically in the last year significantly different from 0.5?

$$\hat{p} - z_\alpha \cdot s_{\hat{p}} < p < \hat{p} + z_\alpha \cdot s_{\hat{p}}$$

$$\hat{p} = \frac{m}{n} = \frac{177}{500} = 0.354 \qquad s_{\hat{p}} = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} = \sqrt{\frac{0.354(1-0.354)}{500}} = 0.021386$$

$z_\alpha = t_\alpha(\infty)$ → $z_{0.05} = t_{0.05}(\infty) = 1.96$ (from 2-tailed t table)

Example CI for the Population Proportion

$$\hat{p} - z_\alpha \cdot s_{\hat{p}} < p < \hat{p} + z_\alpha \cdot s_{\hat{p}}$$

$$0.354 - 1.96(0.021386) < p < 0.354 + 1.96(0.021386)$$

95% CI for proportion of college students who took stimulants non-medically in the last year

$$0.312 < p < 0.396$$

This 95% confidence interval does not contain 0.5, so we are 95% confident that the true population proportion is different from 0.5.

Example CI for the Population Proportion

Hypothesis test for population proportion:

$$H_0 : p = p_0$$

$$z = \frac{sample\ proportion - true\ proportion}{standard\ error\ of\ proportion}$$

$$z = \frac{\hat{p} - p}{s_{\hat{p}}} \qquad s_{\hat{p}} = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

$$\hat{p} - z_\alpha \cdot s_{\hat{p}} < p < \hat{p} + z_\alpha \cdot s_{\hat{p}}$$

Therefore, can construct a CI for true population proportion

Session 17 Lecture Exercise

A drug is purported to reduce fasting blood glucose levels in pre-diabetic patients. In a trial of 1700 subjects, the following results were obtained.

Plasma Glucose with Placebo versus Drug

$$\bar{x}_{pla} = 112.4 \text{ mg/dL} \quad s_{pla} = 4.7 \text{ mg/dL} \quad n = 850$$

$$\bar{x}_{drug} = 109.1 \text{ mg/dL} \quad s_{pla} = 3.9 \text{ mg/dL} \quad n = 850$$

a) Compute the 95% confidence interval for the population difference in mean fasting blood glucose levels between the placebo and drug groups.

b) Is there a statistically significant difference between the two groups at a significance level of 0.05? Answer this question without doing a t test.